

Analysis

PENG

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1 Numbers

1.1 Natural Number system

- This is how mathematics works - it treats its objects abstractly, caring only about what properties the objects have, not what the objects are or what they mean [1].
- It is pointless, at least from a mathematician's standpoint, as to argue about which model is the true one; as long as it obeys all the axioms and does all the right things, that's good enough to do maths.

Axiom 1.1. *Peano axioms:*

1. 0 is a natural number.
2. If n is a natural number, then $n++$ is also a natural number.
3. $n++ \neq 0$
4. $n \neq m \Rightarrow n++ \neq m++$
 $n++ = m++ \Rightarrow n = m$
5. *Principle of mathematical induction:*
Suppose that $P(0)$ is true, and suppose that whenever $P(n)$ is true, $P(n++)$ is also true. Then $P(n)$ is true for every natural number n .

Definition 1.2.

$$\begin{aligned}1 &:= 0++ \\2 &:= 1++ \\3 &:= 2++ \\&\vdots\end{aligned}$$

Definition 1.3. *Addition:*

$$\begin{aligned}0 + m &:= m \\(n++) + m &:= (n + m)++\end{aligned}$$

Lemma 1.4.

$$\begin{aligned}n + 0 &= n \\n + (m++) &= (n + m)++\end{aligned}$$

Proposition 1.5. *Let a, b, c be natural numbers.*

1. $a + b = b + a$
2. $(a + b) + c = a + (b + c)$
3. $a + b = a + c \Rightarrow b = c$

Definition 1.6. *A natural number n is positive iff (if and only if) $n \neq 0$.*

Proposition 1.7. *Let a and b be natural numbers.*

1. *If a is positive, then $a + b$ is positive.*
2. $a + b = 0 \Rightarrow a = b = 0$.
3. *If a is positive, then there exists only one natural number b such that $b++ = a$.*

Proof. Prop. 1.7 (1)

$a + 0 = a$ is positive.

Suppose $a + b$ is positive.

$a + b++ = (a + b)++ \stackrel{\text{Axiom 1.1 (3)}}{\neq} 0$ is positive.

Prop. 1.7 (3)

$1 = 0++$

Suppose $a = b++$

$a++ = (a)++$

□

Definition 1.8. *Let n and m be natural numbers.*

1. $n \geq m$ or $m \leq n$ iff $n = m + a$
2. $n > m$ or $m < n$ iff $n \geq m$ and $n \neq m$

Proposition 1.9. *Let a, b, c be natural numbers.*

1. $a \geq a$
2. *If $a \geq b$ and $b \geq c$, then $a \geq c$*
3. *If $a \geq b$ and $b \geq a$, then $a = b$*
4. $a \geq b$ iff $a + c \geq b + c$
5. $a < b$ iff $a++ \leq b$
6. $a < b$ iff $b = a + d$ for some positive number d

Proof. Prop. 1.9 (5)

$b = a + d$ and $b \neq a$

$d = 0 \Rightarrow b = a$

$d = n++$

$b = a + n++ = (a++) + n \Rightarrow a++ \leq b$

□

Proposition 1.10. *Let a and b be natural numbers. Then exactly one of following statement is true: $a < b$, $a = b$, or $a > b$.*

Proof. If $a < b$ and $a > b \stackrel{\text{Prop. 1.9 (3)}}{\Rightarrow} a = b$ contradiction

□

Proposition 1.11. *Strong principle of induction.*

Let m_0 be a natural number. Suppose that for each $m \geq m_0$, if $P(m')$ is true for all $m_0 \leq m' < m$, then $P(m)$ is also true. Then $P(m)$ is true for all $m \geq m_0$.

Proof.

□

Exercise 1.12. *Principle of backwards induction.*

Exercise 1.13. *Principle of induction starting from the base case n .*

References

[1] Terence Tao. *Analysis I*. Vol. 1. Springer, 2022. ISBN: 978-981-19-7261-4.